

Building Microservices and a CI/CD Pipeline with AWS

AWS Academy Lab Project Outline

Project version

This outline applies to version 1.0 of the Building Microservices and a CI/CD Pipeline with AWS lab project in English.

Description

Throughout various AWS Academy courses, students have completed hands-on labs. These labs included step-by-step guidance to build lab-specific applications by using AWS services and features.

In this project, students are challenged to use AWS services to design and build a microservices and continuous integration and continuous development (CI/CD) solution. First, students break a monolith application into separate microservices, based on functionality and API endpoints. Then, they migrate the application to run on a container-based architecture, which provides the ability for the different components of the application to scale independently. Specific sections of the assignment are meant to challenge students on skills that they have acquired throughout the learning process.

Learning objectives

Upon completion of this project, students will be able to do the following:

- Recognize how a Node.js web application is coded and deployed to run and connect to a relational database where the application data is stored.
- Create an AWS Cloud9 integrated development environment (IDE) and a code repository (repo) in which to store the application code.
- Split the functionality of a monolithic application into separate containerized microservices.
- Use a container registry to store and version control containerized microservice Docker images.
- Create a code repository to store microservice source code and CI/CD deployment assets.
- Create a serverless cluster to fulfill cost optimization and scalability solution requirements.
- Configure an Application Load Balancer and multiple target groups to route traffic between microservices.
- Create a code pipeline to deploy microservices containers to a blue/green cluster deployment.
- Use the code pipeline and code repository for CI/CD by iterating on the application design.

Duration

The duration will vary depending on how educators integrate this project into course activities and can be from one week to one semester. The project includes the following:

- Planning activities, such as creating an architectural diagram and a cost estimate of what students will build.
- Implementation time to build the solution. Building the solution in the lab environment is anticipated to take a student 10–14 hours.
- Post-build activities, such as creating a presentation or recording a demo to share the solution with the educator and other students. Students might also create an AWS CloudFormation template to re-create the solution later in another account.

To spread out the work across a semester, educators can have students work on planning activities for a few weeks, implementation for a few weeks, and then have students present their work. Educators can also choose to have students plan, implement, and present their solutions within a single week or two.

Intended audience

This project is intended for students who attend AWS Academy member institutions, seek to gain work-based experience, and are interested in developing a career as a software development engineer (SDE), app developer, or cloud support engineer.

Student prerequisites

To ensure success with this project, students are strongly recommended to have the following:

- Completed the AWS Academy Cloud Foundations course or have equivalent experience
- Completed the AWS Academy Cloud Architecting, AWS Academy Cloud Operations, or AWS Academy Cloud Developing course or have equivalent experience
- Worked with core AWS services; for example, compute, storage, networking, and database services
- Knowledge of architectural best practices, such as high availability, scalability, and security
- Knowledge of Linux Bash commands (Note: This experience would provide an advantage but is not mandatory.)

Educator prerequisites

This project does not have any prerequisites for educators. However, prior to facilitating the project, educators are recommended to complete this project; the AWS Academy Cloud Foundations course; and the AWS Academy Cloud Architecting, AWS Academy Cloud Operations, or AWS Academy Cloud Developing course. Educators are also recommended to pass the AWS Certified Cloud Practitioner exam, and the AWS Certified Solutions Architect – Associate exam or AWS Certified Developer – Associate exam.

Delivery methods

Learning materials are provided to support in-person or online synchronous delivery. Educators can determine the level of scaffolding to provide, such as using a guided approach or having students complete the project with limited instructor support.

Learning resources

The following learning resources are provided for the project:

- Instructor guide
- Grading rubric
- Lab instructions
- Lab environment
- Knowledge check
- Showcase presentation template

Note: To access the instructor guide, download and extract the .zip file from the educator files folder. From the extracted folder, open the IG.html file.